Pathways to Better Postpartum Sleep: Mediators of CBT-I and Responsive Bassinet Effects in a Randomised Trial

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INTRODUCTION

Perinatal insomnia is prevalent and associated with adverse maternal and child health outcomes.
How **interventions improve maternal postpartum insomnia** is not well explored**.**

Secondary analysis testing:

- Unhelpful beliefs about sleep (DBAS) change as a mediator of cognitive behavioural therapy for insomnia (CBT-I) effects.
- Infant and maternal sleep as mediators of responsive bassinet effects.

METHODS

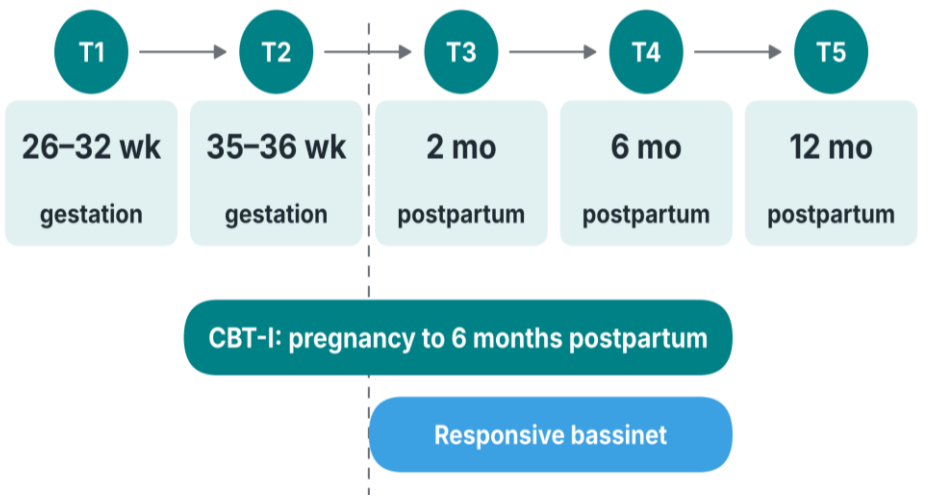
Participants 127 first-time mothers (age = 32.62 ± 3.49) with insomnia symptoms (insomnia severity index [ISI] > 7) at late pregnancy.

Enrolled in **SMILE RCT** comparing effects of two sleep interventions, vs control.

Interventions

- Therapist-assisted **CBT-I group** (n = 42)
- Smart **Responsive Bassinet group** (n = 44)
- Sleep Hygiene **Control group** (n = 41)

Survey and Intervention Timepoints



Outcome Average postpartum insomnia.

Analysis Mediation analyses, bootstrapped 20,000 times. Direct, indirect and total effects.

RESULTS

CBT-I Mediation

- Total effect (c): CBT-I associated with **lower postpartum ISI**
- CBT-I associated with a **greater reduction in DBAS**
- Indirect effect: **DBAS change not supported as a mediator**

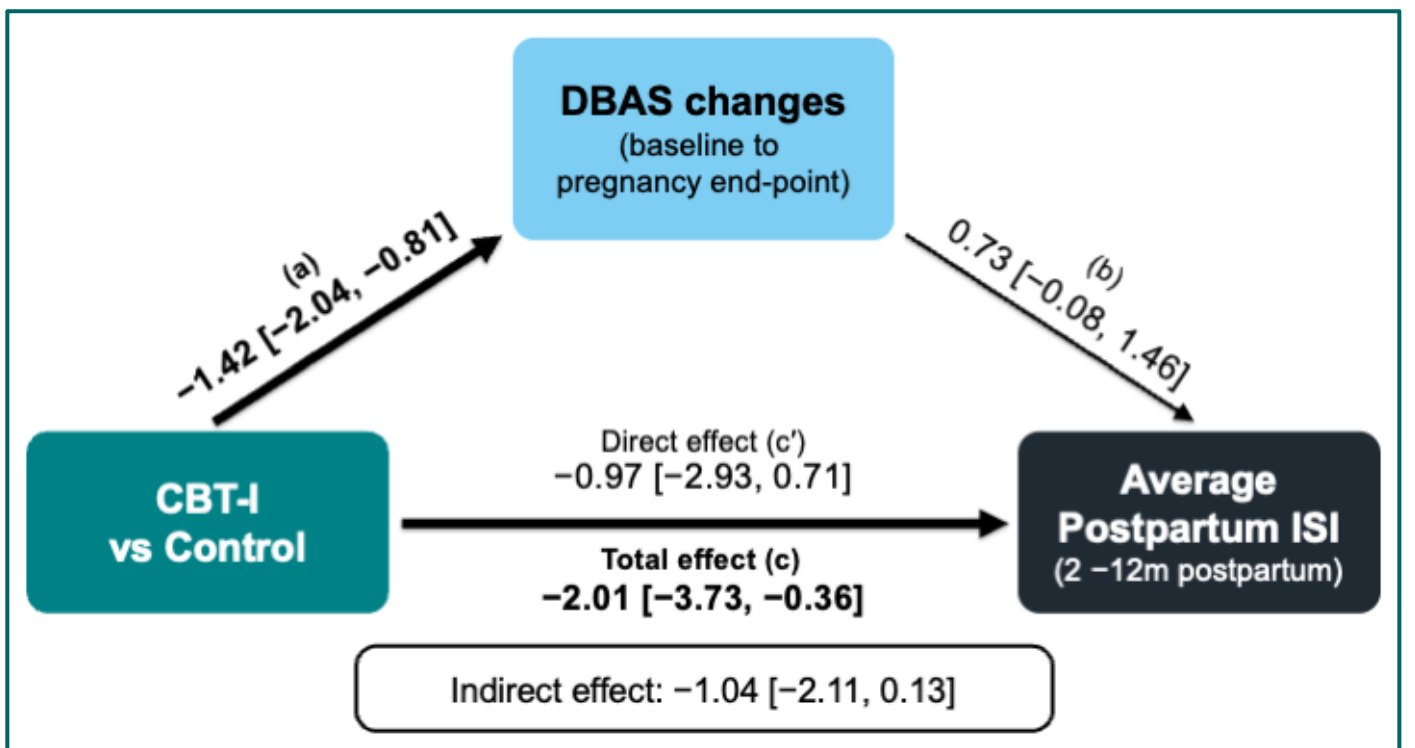


Figure 1. DBAS change as a mediator of CBT-I effects on postpartum ISI

Responsive Bassinet Serial Mediation

- Total effect (f) of **RB on postpartum ISI not supported**
- Indirect effect (d*c): **RB associated with lower ISI via longer maternal TST**
- **Infant SOL not supported as a mediator**

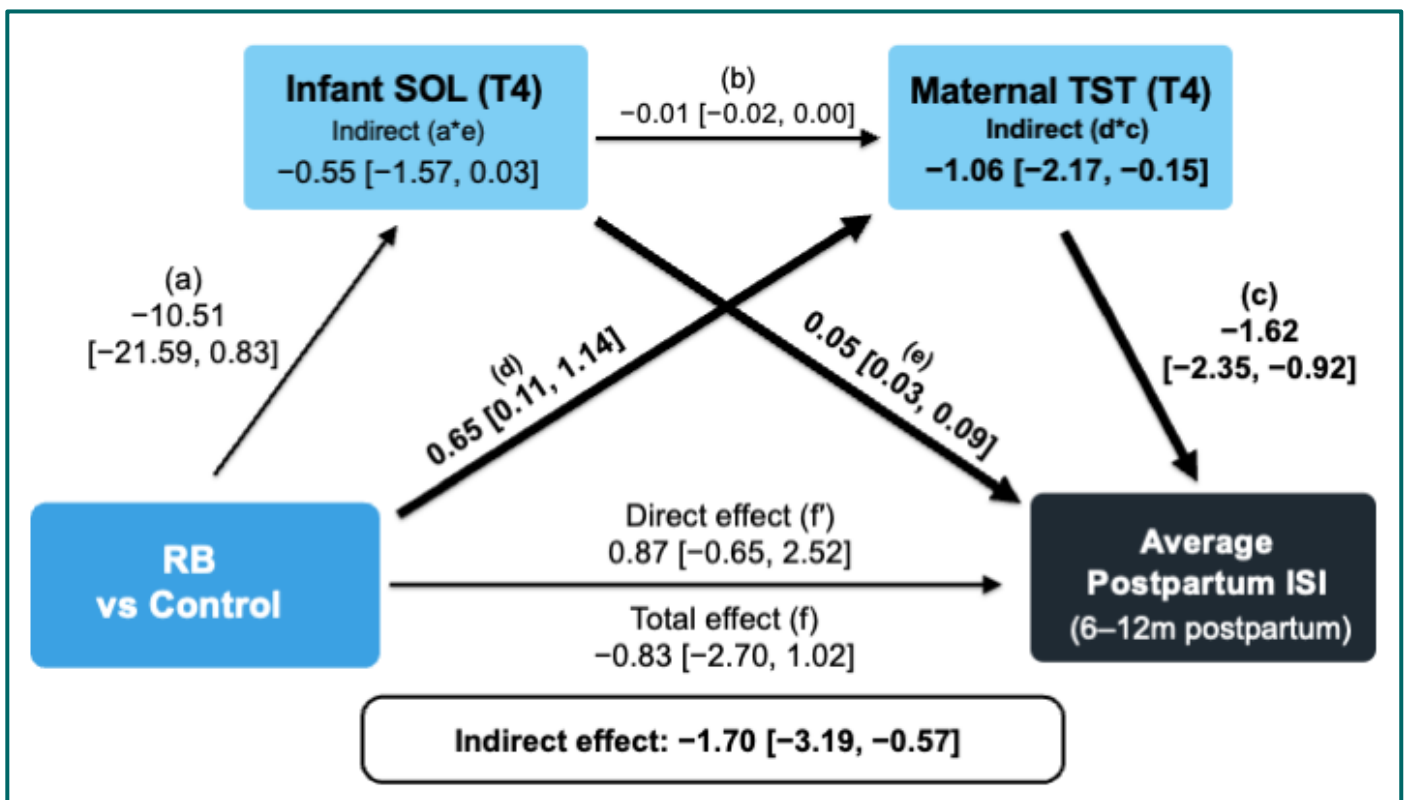


Figure 2. Infant SOL and maternal TST as serial mediators of responsive bassinet effects on postpartum ISI

CONCLUSIONS AND IMPLICATIONS

- **CBT-I** (vs control) associated with **lower postpartum ISI**. **DBAS** change **not supported** as mediator → likely underpowered.
- The **responsive bassinet** (vs control) associated with an **indirect effect**, via **maternal sleep duration** rather than infant sleep.
- The benefit of **CBT-I** may occur through mechanisms other than DBAS, or the DBAS measure may not capture the relevant change. Future work should examine mechanisms, including sleep effort, arousal or pre-sleep worry, in **larger samples**.
- **Technology-assisted soothing** may be one potential strategy for **supporting maternal sleep**.